AWANTIKA SRIVASTAVA

Data Scientist | Machine Learning Engineer | AI Engineer

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PROFILE SUMMARY

Data Scientist and Machine Learning Engineer with **2+ years of experience building machine learning and AI solutions in enterprise environments**. Strong expertise in **Python, Deep Learning, PyTorch, ETL pipelines, and feature engineering** for structured and unstructured data. Experienced in developing **predictive models, anomaly detection systems, and data processing pipelines**. Skilled in **GenAI pipelines, document extraction, model evaluation, and MLOps workflows** using tools such as **Docker, MLflow, and cloud platforms**. Proven ability to transform complex datasets into **actionable insights and scalable AI solutions**.

CORE TECHNICAL SKILLS

- **Programming & Data Handling:** Python, Numpy, Pandas, Matplotlib, Scikit-learn, C++, SQL
- **Statistics & Mathematics:** EDA, Statistical Analysis & Modeling, Hypothesis Testing, A/B Testing, Probability, Linear Optimization
- **Machine Learning:** Supervised & Unsupervised Learning, Regression, Classification, Clustering, Random Forest, Decision Trees, SVM, KNN, K-Means, XGBoost, Hyperparameter Tuning, Feature Engineering, Anomaly Detection
- **Deep Learning:** Neural Networks, CNN, RNN, LSTM, BERT, Vision Transformers, Self-Attention, Cross-Attention, Sequence Modeling, TensorFlow, PyTorch, Keras, Time-Series Forecasting, ARIMA, SARIMA
- **Computer Vision**: Image Classification & Preprocessing, Object detection (YOLO, SSD, MobileNet, ResNet), Feature Extraction, Paddle OCR
- **NLP**: Text preprocessing, Tokenization, Chunkings, Transformers, Sentiment Analysis, Semantic Search, Transformer-based models, NLTK, RAG Pipeline, GenAI
- **MLOps & Tools:** Docker, MLflow, Git, CI/CD, Linux, Git, Jupyter Notebook
- **Databases & Tools:** MySQL, PostgreSQL
- **Visualization:** PowerBI, Matplotlib, Seaborn

EXPERIENCE

Data Scientist | PPS International Pvt. Ltd. **January 2024-Present**

- Developed **machine learning models using Python and PyTorch** for predictive analytics and anomaly detection tasks.
- Built scalable **ETL pipelines to process structured and unstructured datasets** for model training and analytics.
- Applied **feature engineering and statistical analysis techniques** to improve model accuracy and predictive performance.
- Implemented **supervised and unsupervised learning algorithms** to identify patterns and insights from large datasets.
- Designed **model evaluation frameworks and validation pipelines** to ensure reliability and performance of deployed models.
- Built **data preprocessing pipelines** using Pandas and NumPy to clean and transform large datasets.
- Developed **AI pipelines for document processing and information extraction from unstructured data sources**.
- Worked with engineering teams to **deploy machine learning models using containerized environments (Docker)**.
- Collaborated with cross-functional teams to translate business requirements into **data science solutions and analytical insights**.

PROJECTS

Railway Driver Assistance System (RDAS) | Enterprise ML Project

- Designed and **deployed a real-time computer vision**–based ML system to detect unsafe driver behaviors from continuous video streams.
- Trained and **optimized CNN-based** object detection **models (SSD MobileNet** architecture) to perform real-time inference on video data.
- **Implemented end-to-end ML pipelines** covering data ingestion, preprocessing, model training, evaluation, and production inference.
- Achieved **20–25 FPS** real-time processing with **<150 ms** inference latency by optimizing models for **deployment**.
- Built and integrated **a Flask-based** web **interface** to **visualize** detections and automatically recorded 30-second event clips, reducing manual **review** effort.

LLM & RAG-Based AI Chatbot | Applied ML Project

- Built a Lightweight **LLM-Powered chatbot** using **TinyLLaMA** to answer user queries over content.
- Implemented a **Retrieval-Augmented Generation (RAG) pipeline** to retrieve relevant resume sections for contextual question answering.
- Selected **TinyLLaMA** to ensure **low memory footprint and fast inference**, making the solution suitable for resource-constrained environments.
- Deployed the chatbot as an interactive **Streamlit web application** for real-time user interaction.

YouTube Comments Sentiment Analyzer | Link -https://youtube-ai-analyzer-ndzqo6r2mepjrtsdjmwaxl.streamlit.app/

- Deployed transformer-based **NLP models** as **production-ready** services with **REST APIs**.
- **Fine-tuned** and served a **DistilBERT**-based **sentiment** classification model for large-scale text inference.
- **Built** and deployed an interactive **streamlit web application** to perform real-time **sentiment analysis** on YouTube comments.
- **Processed** high-volume text **data** with sub-second inference **latency** for real-time sentiment analysis.

CERTIFICATION

- IBM Data Science & AI Certification
- AWS Generative AI with Large Language Models
- OpenCV Computer Vision Certification

EDUCATION

IMS Engineering College, Ghaziabad **September - 2020**
Bachelor of Technology